

EC3214 — International Finance

Summer 2025 Examination · Paper 1 · Full Marking Scheme

Module	EC3214 — International Finance
Internal Examiner	Dr Hugh Farrell
Institution	University College Cork
Session	Summer 2025, Paper 1
Duration	1.5 hours
Structure	Section A — Long Questions: attempt 2 of 3 (100% of the paper). All questions carry equal weight.
Materials	Non-programmable calculator permitted

Coverage note. This scheme answers **all three** Section A questions in full, since you'll choose your best two under exam conditions.

Own-figure rule. Where a question part carries a numerical result forward from an earlier step, an answer that applies the correct method consistently from an earlier arithmetic slip still earns the marks available for the later steps.

Source basis. A1, A2(a)-(b), and the monetary-model material in A3(b) follow the module's own lecture slides and assigned readings (including the Pakko and Pollard Federal Reserve Bank of St. Louis reading on PPP and the law of one price) closely. Three parts of this paper turn on standard international finance theory that the uploaded course material referenced but didn't fully spell out: the fixed-exchange-rate Mundell-Fleming case (the course's own reading list notes the dedicated PDF for this topic wasn't available), Optimum Currency Area theory (A3(a)), and the first-generation currency crisis model (A3(c)). These follow the standard treatment (Mundell, 1961; Krugman, 1979) and are flagged at each question.

Section A · Question 1

The law of one price, PPP and the real exchange rate, and interest rate parity conditions — 100 marks

A1(a)

[30 marks]

You find that the price of a one-hour lawyer consultation is €100 in Cork and £70 in London. If the GBP/EUR exchange rate is £1 = €1.21, is this a violation of the Law of One Price? Explain why or why not.

A1(b)

[20 marks]

Absolute Purchasing Power Parity (PPP) is given as: $P = SP^*$. If the above holds true, what does this mean for the Real Exchange Rate? Can this be used as a test of PPP?

A1(c)

[25 marks]

Uncovered Interest Rate Parity (URIP) can be given mathematically as: $r = r^* + \Delta S^e$. Explain why we must make the assumption that agents are risk neutral for the above to hold.

A1(d)

[25 marks]

With Covered Interest Parity (CIRP) we do not need to assume agents are risk neutral. Explain why this is the case and show how the CIRP equation differs from the UIRP equation above.

FULL MODEL ANSWER

Converting to a common currency: $£70 \times 1.21 = €84.70$, compared with the €100 Cork price — a gap of €15.30, with London appearing considerably cheaper once converted.

Is this a violation of the Law of One Price? No. The law of one price applies specifically to **tradable goods** — items that can be bought in one location and physically resold in another to arbitrage away any price gap. A one-hour lawyer consultation is a **non-tradable service**: it must be consumed at the location where it is produced, and there is no way to "buy" the London consultation and resell it to a client in Cork. Because the arbitrage mechanism that enforces the law of one price for tradable goods (buy where cheap, sell where dear, until the gap closes) is structurally unavailable for this kind of service, the observed price gap is not evidence of any market failure or

mispricing — it simply reflects that this good sits entirely outside the scope of goods the law was ever meant to describe.

What could still explain the gap, without invoking arbitrage: genuine differences in the underlying cost of providing the service — commercial rent, professional insurance costs, local wage levels, or the intensity of local competition among lawyers — can differ substantially between Cork and London without implying anything is "wrong" or exploitable about the pricing.

Mark Allocation — Part (a) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly converts the London price to EUR and identifies the gap AND correctly concludes this is NOT a LOP violation AND correctly explains why — because the good in question (a personal service) is non-tradable and therefore cannot be arbitrated.
15-22	15-22	Correct conversion and correct "not a violation" conclusion, but the explanation doesn't clearly invoke non-tradability/the impossibility of arbitrage as the reason.
8-14	8-14	Correct conversion, but incorrectly concludes this IS a LOP violation, or gives no clear reasoning for the conclusion reached.
0-7	0-7	Currency conversion incorrect, or no credible reasoning given.

What absolute PPP implies for the real exchange rate: the real exchange rate is defined as $Q = SP^*/P$ (the nominal exchange rate adjusted for relative price levels). If absolute PPP ($P = SP^*$) holds exactly, then by direct substitution $Q = SP^*/P = SP^*/(SP^*) = 1$ — the real exchange rate is identically equal to 1, by construction, whenever absolute PPP holds. This is not a coincidental result; it is exactly what absolute PPP *means*: that a given basket of goods costs the same everywhere once converted to a common currency, so there is no "real" difference in purchasing power left over once the nominal exchange rate is accounted for.

Can this be used as a test of PPP? Not directly, and only with important caveats. The equation $Q = 1$ is testable in principle, but price levels P and P^* are not observed as absolute prices — they are measured as **index numbers** (e.g. CPI), which are constructed relative to an arbitrarily chosen base period (index = 100 in some reference year) and are built from baskets of goods that differ in composition and weighting across countries. Because of this, the *level* of Q computed from real-world CPI data has no natural interpretation as "1" — the index-number base periods and basket weights are arbitrary choices that don't correspond to any true common unit across countries. What *can* be meaningfully tested is whether Q is **constant over time** (the relative form of PPP, or equivalently, whether Q is stationary/mean-reverting) — since changes in an index number over time within a single country are meaningful, even though the index's level isn't comparable across countries.

EXAM TECHNIQUE

This is a classic and important distinction in the literature (see, e.g., Pakko and Pollard's discussion of PPP testing in the module's assigned reading): absolute PPP (" $Q=1$ ") is a clean theoretical statement, but testing it directly requires a common, identically-constructed basket of goods across countries (the Penn World Table approach) — with standard national CPI data, only relative PPP (constancy of Q over time) is genuinely testable.

Mark Allocation — Part (b) (20 marks)

Tier	Marks	What separates this tier
15-20	15-20	Correctly derives $Q=1$ from $P=SP^*$ by substitution AND correctly explains that $Q=1$ cannot be directly tested with standard index-number price data due to arbitrary base periods/basket composition AND correctly identifies that relative PPP (constancy of Q over time) is the more testable implication.
10-14	10-14	Correctly derives $Q=1$, but the index-number testing caveat is missing or only vaguely gestured at.
5-9	5-9	States $Q=1$ without deriving it from the given equation, or without addressing the testability question at all.
0-4	0-4	No credible answer given.

Why UIP needs risk neutrality: UIP as stated, $r = r^* + \Delta S^e$, says the domestic interest rate exactly equals the foreign interest rate plus the **expected** rate of currency depreciation — with no additional term. But holding a foreign-currency asset uncovered (with no forward contract locking in the future conversion rate) exposes the investor to genuine **exchange rate risk**: the actual future spot rate could differ from what's currently expected. A **risk-averse** investor would only be willing to bear that uncertainty in exchange for compensation — a risk premium — meaning the equation would need an extra term: $r = r^* + \Delta S^e + \text{risk premium}$, with the domestic rate needing to exceed the foreign rate plus expected depreciation by enough to compensate for the risk actually borne. It is only under **risk neutrality** — where investors care solely about the *expected value* of their return and are entirely indifferent to the variance/uncertainty around it — that this risk-premium term drops out entirely, leaving the clean equality in the equation as given.

Mark Allocation — Part (c) (25 marks)

Tier	Marks	What separates this tier
19-25	19-25	Correctly identifies that holding an uncovered foreign asset carries exchange rate risk AND correctly explains that a risk-averse investor would demand a risk premium, requiring an additional term in the equation AND correctly explains that risk neutrality specifically is what removes this term, since risk-neutral investors care only about expected value.
13-18	13-18	Correctly identifies exchange rate risk as the reason a risk premium might be needed, but doesn't precisely explain why risk neutrality (rather than some other assumption) is what removes it.
7-12	7-12	Vague reference to "risk" without explaining the specific mechanism (uncovered position, risk premium, risk-neutral indifference to variance).
0-6	0-6	No credible explanation given.

Why CIRP doesn't need risk neutrality: covered interest parity uses a **forward contract**, agreed today, to lock in the exchange rate at which foreign-currency proceeds will be converted back at maturity. Because the conversion rate is contracted in advance, there is **no exchange rate risk left in the transaction at all** — the outcome is certain from the moment the trade is put on, regardless of how the actual spot rate later moves. Since there is no risk to be compensated for, the condition holds purely as a **no-arbitrage identity** among currently observable, contractible prices — it is a mechanical consequence of ruling out risk-free profit opportunities, and holds regardless of whether market participants are risk-averse, risk-neutral, or risk-seeking.

How the CIRP equation differs from UIRP: the key difference is which future exchange rate enters the equation:

UIRP: $r = r^* + \Delta S^e$ — uses the **expected**, uncertain future spot rate (ΔS^e)

CIRP: $r = r^* + (F - S)/S$ — uses the **actual, currently observable, contracted forward rate** (F)

Why this distinction is exactly the point: UIRP involves a genuinely *subjective, forward-looking expectation* (ΔS^e) about an uncertain future outcome — this is why it's a behavioural/expectational relationship that can fail to hold in practice (investors' expectations can be wrong, or a risk premium can drive a wedge between the two sides). CIRP instead involves only prices that are *already fixed and known today* — the spot rate S, the forward rate F, and the two interest rates — making it a mechanical identity that must hold (up to transaction costs) or a genuine, risk-free arbitrage opportunity would exist, exactly as demonstrated numerically in the EC3213 Winter 2025 scheme, Q. A1(a).

Mark Allocation — Part (d) (25 marks)

Tier	Marks	What separates this tier
19-25	19-25	Correctly explains that the forward contract eliminates exchange rate risk entirely, making CIRP a pure no-arbitrage identity AND correctly writes the CIRP equation using the forward premium $(F - S)/S$ AND correctly contrasts this with UIRP's use of the expected, uncertain ΔS^e .
13-18	13-18	Correctly explains why CIRP doesn't need risk neutrality, but the equation itself is stated incorrectly or the F vs. ΔS^e distinction isn't made explicit.
7-12	7-12	States that CIRP "is different from UIRP" without a clear explanation of the risk mechanism or the equation difference.
0-6	0-6	No credible explanation given.

Question A1 Total: 100 Marks

Section A · Question 2

Cross exchange rates, the monetary model, and Mundell-Fleming under fixed rates — 100 marks

A2(a)

[20 marks]

Consider the following exchange rates: $\text{USD/JPY} = 152.15$, $\text{USD/GBP} = 0.81$. Find the GBP/JPY exchange rate.

A2(b)

[40 marks]

In the Monetary Model of Exchange Rate Determination, it is assumed that absolute PPP holds and that the following is true: $M_o^s = kPy$. If the foreign price level increases, what variables must change in order for equilibrium to be restored in a floating exchange rate regime? Will there be any real effect on the domestic economy? Explain your answer and illustrate with the aid of a diagram.

A2(c)

[40 marks]

Outline the effect of a monetary expansion in a fixed exchange rate regime in the Mundell-Fleming Model with the aid of a diagram.

FULL MODEL ANSWER

Setting up the cross rate: under the standard quote convention, USD/JPY = 152.15 means 152.15 JPY per 1 USD, and USD/GBP = 0.81 means 0.81 GBP per 1 USD. To find GBP/JPY (JPY per 1 GBP), divide the two USD-based quotes so the USD terms cancel:

Step	Calculation	Result
USD/JPY	given	152.15 JPY per USD
USD/GBP	given	0.81 GBP per USD
Implied GBP/USD	$1 \div 0.81$	1.2346 USD per GBP
GBP/JPY	$152.15 \div 0.81$ (equivalently, 1.2346×152.15)	187.84 JPY per GBP

EXAM TECHNIQUE

The key mechanical step is dividing the two USD-quoted rates (rather than multiplying) so that the common USD term cancels out of the cross rate — multiplying them instead would leave USD² in the units and give a meaningless result.

Mark Allocation — Part (a) (20 marks)

Tier	Marks	What separates this tier
15-20	15-20	Correctly identifies that the USD terms must cancel via division AND arrives at GBP/JPY = 187.84 (or the equivalent inverse quote, correctly labelled).
10-14	10-14	Correct method (dividing rather than multiplying) but an arithmetic error in the final figure.
5-9	5-9	Incorrect method (e.g. multiplying the two rates) but with the calculation carried through consistently.
0-4	0-4	No coherent cross-rate calculation attempted.

Setting up the monetary model: money market equilibrium requires $M_o^s = kPy$ — the money supply equals money demand, which is proportional to the price level P and real income y. Rearranging for the price level: $P = M_o^s / (ky)$. Absolute PPP (see Q. A1(b)) then links the domestic price level to the exchange rate: $S = P/P^*$.

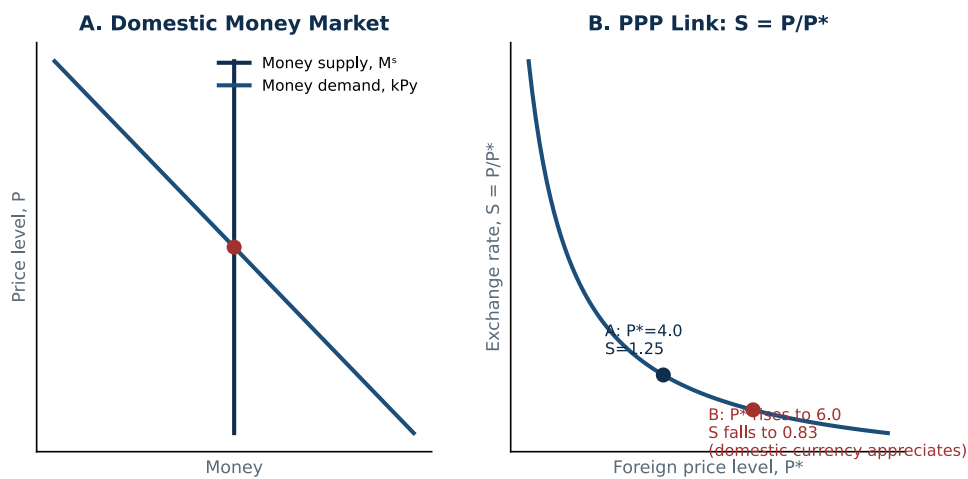


Fig. 1 — Panel A: the domestic money market fixes P given M_o^s , k , and y . Panel B: PPP maps a given P into $S = P/P^*$; when P^* rises (from panel B's point A to B), S falls — the domestic currency appreciates.

What must change to restore equilibrium: if the foreign price level P^* rises, and **nothing on the domestic side changes** (M_o^s , k , and y are all unaffected by a purely foreign price shock), then P itself cannot change — it remains pinned down entirely by the unchanged domestic money market equation $P = M_o^s/(ky)$. The **only** variable left that can adjust to restore the PPP equation $S = P/P^*$ is the **nominal exchange rate S itself**: with P fixed and P^* higher, S must **fall** — the domestic currency **appreciates**.

Will there be any real effect on the domestic economy? No. This is one of the defining features of the pure monetary model: because absolute PPP is assumed to hold continuously (prices are fully flexible, with no stickiness at all), and because output y is determined independently at its full-employment level, a purely foreign monetary/price disturbance is absorbed **entirely** by the nominal exchange rate, with no knock-on effect on domestic output, employment, or real income. Money — and, by extension, the foreign price level shock transmitted through it — is fully **neutral** in this model: only nominal variables (P , S) move; real variables (y) do not.

EXAM TECHNIQUE

This is exactly why the module's own material flags the pure monetary model's biggest shortcoming: because PPP rarely holds well in the short or medium run in reality (see Q. A1(b)'s testing discussion), this "no real effects, instant full adjustment via S " prediction is unrealistic over short horizons — which is precisely the gap the Dornbusch overshooting model (with short-run price stickiness) was built to address.

Mark Allocation — Part (b) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Correctly sets up $P = M_o^s/(ky)$ and $S = P/P^*$ AND correctly concludes that S alone must adjust (fall) to restore equilibrium, since domestic variables are unaffected by a foreign shock AND correctly concludes there are no real effects, with the money-neutrality reasoning explained AND the diagram correctly shows the mechanism.
21-30	21-30	Correctly concludes S must fall and that there are no real effects, but the reasoning (why P can't adjust; why output is unaffected) is only partially explained.
11-20	11-20	States that the exchange rate changes, but the direction is incorrect or the "no real effects" conclusion is missing or wrong.
0-10	0-10	No credible answer given.

Setting up Mundell-Fleming with perfect capital mobility: under a fixed exchange rate regime with high (here, perfect) capital mobility, the balance-of-payments (BP) curve is **horizontal** at the world interest rate — any domestic interest rate below the world rate triggers unlimited capital outflow, and any rate above it triggers unlimited inflow, since capital can move freely and instantly to wherever it earns the higher return.

Monetary Expansion Under Fixed Exchange Rates (Mundell-Fleming)

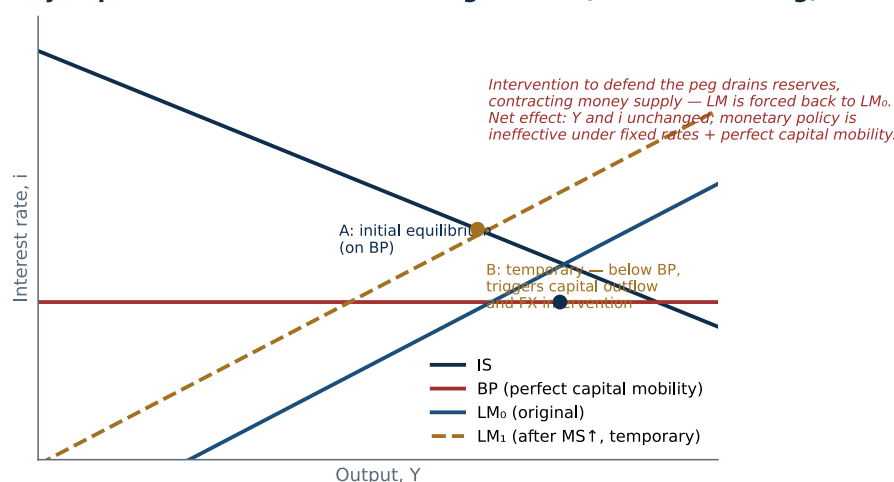


Fig. 2 — A monetary expansion shifts LM right (LM_0 to LM_1), temporarily moving the economy to point B below the BP line. The resulting capital outflow forces central bank intervention to defend the peg, draining reserves and money supply until LM returns to LM_0 — output and the interest rate end up unchanged.

The mechanism, step by step: a monetary expansion initially shifts LM to the right ($LM_0 \rightarrow LM_1$), which — taken on its own — would lower the domestic interest rate and raise output, moving the economy to point B, **below** the BP line. But a domestic interest rate below the world rate is exactly what triggers capital **outflow** under perfect capital mobility: investors sell domestic-currency assets for the higher-yielding foreign ones. Because the exchange rate is **fixed**, the central bank is committed to supplying whatever foreign currency is demanded at the pegged rate — meaning it must sell foreign reserves and buy back domestic currency to defend the peg. This intervention directly **contracts the domestic money supply**, which shifts LM back to the **left** — and this process continues until LM returns all the way to its original position, LM_0 , restoring the interest rate to the world level and closing off the capital outflow.

The result: monetary policy is ineffective under fixed rates with perfect capital mobility. output and the interest rate end up completely unchanged from where they started — the entire monetary expansion is automatically and fully reversed by the foreign exchange intervention required to defend the peg. This is one of the sharpest results in open-economy macroeconomics: a fixed exchange rate commitment, combined with free capital mobility, removes monetary policy as an independent tool entirely — a country can have at most two of "fixed exchange rate," "free capital mobility," and "independent monetary policy" simultaneously (the open-economy "trilemma").

Mark Allocation — Part (c) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Correctly sets up the horizontal BP curve under perfect capital mobility AND correctly traces the full mechanism (LM shifts right $\rightarrow$ below BP $\rightarrow$ capital outflow $\rightarrow$ FX intervention to defend the peg $\rightarrow$ money supply contracts $\rightarrow$ LM shifts back to LM_0) AND correctly concludes monetary

policy is fully ineffective, ideally connecting this to the open-economy trilemma.

21-30	21-30	Correctly identifies that the monetary expansion is reversed by FX intervention, but the step-by-step mechanism (particularly the capital-outflow trigger) isn't fully traced through.
11-20	11-20	Diagram or general description given, but the "ineffective" conclusion is missing or the mechanism is only vaguely gestured at.
0-10	0-10	No credible answer given.

Question A2 Total: 100 Marks

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Q2

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Section A · Question 3

Optimum Currency Areas, comparing exchange rate models, and the first-generation currency crisis model — 100 marks

A3(a)

[30 marks]

What is meant by an Optimum Currency Area (OCA)? Outline the criteria which determines whether an area can be considered an OCA.

A3(b)

[30 marks]

Discuss the assumptions underlying the Mundell-Fleming model of exchange rate determination and explain how they differ from the assumptions underlying the Monetary Model.

A3(c)

[40 marks]

Describe the First-Generation Model of Currency Crises. At what point will the fixed exchange rate regime collapse?

FULL MODEL ANSWER

EXAM TECHNIQUE

Optimum Currency Area theory (Mundell, 1961) wasn't set out in dedicated lecture slides in the uploaded course material, though it's directly relevant to the module's Eurozone context. The definition and criteria below follow the standard treatment.

What an Optimum Currency Area is: a geographic region for which it is economically optimal — in the sense of maximising the net benefit relative to each member retaining its own independent currency — to share a single

currency and a single monetary policy, rather than each constituent area having its own currency and central bank. The core trade-off: sharing a currency eliminates exchange-rate uncertainty and transaction costs within the area, but it also means giving up the ability to use an independent monetary policy or exchange rate adjustment to respond to shocks that hit different parts of the area differently.

The criteria

- **Labour mobility:** if workers can move freely and easily between regions within the currency area, a region hit by an adverse shock can adjust via out-migration to regions doing better, substituting for the exchange-rate devaluation that region could otherwise no longer use on its own.
- **Wage and price flexibility:** if wages and prices adjust quickly, a region facing reduced competitiveness can regain it through falling relative wages/prices (an "internal devaluation") instead of a nominal exchange-rate depreciation.
- **Openness and trade integration:** the more open and integrated the region's economies are with each other, the larger the transaction-cost and exchange-rate-uncertainty savings from a shared currency, and the smaller the independent monetary policy would have mattered anyway.
- **Fiscal transfers / risk-sharing mechanisms:** a central fiscal capacity that automatically transfers resources toward regions hit by an adverse shock can substitute for the lost exchange-rate and monetary-policy tools, cushioning the region without requiring it to regain competitiveness on its own.
- **Similarity of economic shocks / business cycle synchronisation:** if the constituent regions tend to be hit by similar shocks at similar times, a single monetary policy set for the area as a whole will tend to suit every region reasonably well, since there is little need for region-specific responses in the first place.

Why this matters for the module's own context: these criteria are the standard lens through which the Eurozone's own design is assessed — labour mobility across EU member states is real but limited (by language and institutional barriers relative to, say, mobility across US states), and the Eurozone historically lacked a large central fiscal transfer mechanism, which is a core part of the long-running academic debate about whether the Eurozone constitutes a genuine OCA.

Mark Allocation — Part (a) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correct definition of an OCA as a cost-benefit trade-off (currency-union benefits vs. loss of independent monetary/exchange-rate adjustment) AND at least four distinct criteria correctly named and explained AND the criteria are connected back to why they substitute for the lost adjustment tools.
15-22	15-22	Correct definition with at least three criteria named and briefly explained.
8-14	8-14	Vague definition of an OCA without a clear cost-benefit framing, or fewer than three criteria given.
0-7	0-7	No credible definition or criteria given.

Mundell-Fleming's assumptions: a **short-run**, Keynesian framework: prices are assumed **sticky/fixed** in the short run, so output can deviate from its full-employment level in response to demand shocks — exactly the mechanism that let monetary policy move both output and the interest rate in Q. A2(c) before capital flows and FX intervention took over. It models equilibrium jointly across the **goods market** (IS), the **money market** (LM), and the **balance of payments** (BP), with the degree of capital mobility a central, explicit

parameter of the model (from zero capital mobility to the perfect mobility case used in Q. A2(c)).

The Monetary Model's assumptions: a **longer-run**, classical framework: prices are assumed **fully flexible**, so absolute PPP holds continuously (see Q. A2(b)) and output sits permanently at its full-employment level, determined independently of monetary factors — money is **neutral**. There is no separate goods-market (IS) equilibrium condition at all; the exchange rate is determined purely by the **money market** ($M_0^s = kPy$ in each country) together with the PPP link, with no role for capital mobility or interest-rate-driven capital flows in the basic version of the model.

	Mundell-Fleming	Monetary Model
Time horizon	Short run	Long run
Prices	Sticky/fixed	Fully flexible
Output	Can deviate from full employment (demand-determined)	Fixed at full employment (supply-determined)
PPP	Not assumed to hold	Assumed to hold continuously (absolute PPP)
Markets modelled	Goods (IS) + money (LM) + balance of payments (BP)	Money market only, linked via PPP
Role of capital mobility	Central, explicit — determines the shape of BP	No explicit role in the basic version
Exchange rate driven by	Interest rates, output, capital flows	Relative money supplies and price levels

The underlying reason for the difference: the two models are best understood as answering different questions over different time horizons rather than as competing claims about the same thing — Mundell-Fleming is built to analyse how policy affects the economy *before* prices have had time to adjust (hence its focus on real output effects), while the Monetary Model describes where the exchange rate settles *once* prices have fully adjusted. This is exactly the gap the Dornbusch overshooting model (see the module's own material) was built to bridge, by combining Mundell-Fleming-style short-run price stickiness with the Monetary Model's long-run flexible-price PPP outcome in a single, unified framework.

Mark Allocation — Part (b) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Both models' core assumptions correctly stated (sticky prices/real output effects vs. flexible prices/PPP/full employment) AND at least three specific points of contrast correctly drawn AND the underlying reason for the difference (short-run vs. long-run focus) is explained.
15-22	15-22	Both models' assumptions correctly stated with some contrast drawn, but the underlying short-run/long-run rationale for the difference is not explained.
8-14	8-14	Only one model's assumptions correctly and fully stated.
0-7	0-7	Neither model's assumptions correctly stated.

EXAM TECHNIQUE

The first-generation currency crisis model (Krugman, 1979) wasn't set out with its full formal derivation in the uploaded course material, though the module's own readings

extensively document the real-world crises (the 1990s ERM, Mexican peso, and Asian currency crises) this model was built to explain. The mechanics below follow the standard treatment of Krugman's model.

The setup: a government runs a persistent fiscal deficit that it finances by expanding domestic credit — effectively printing money to cover the shortfall — while simultaneously trying to maintain a **fixed exchange rate**. Domestic money demand grows only slowly (in line with the economy), but domestic credit expansion is growing faster than that — so to keep the money supply consistent with the fixed peg, the excess money creation must be continuously offset by a **gradual loss of foreign exchange reserves**, as the central bank sells reserves to buy back the excess domestic currency being created by the ongoing credit expansion.

First-Generation Currency Crisis: Reserve Depletion and the Speculative Attack

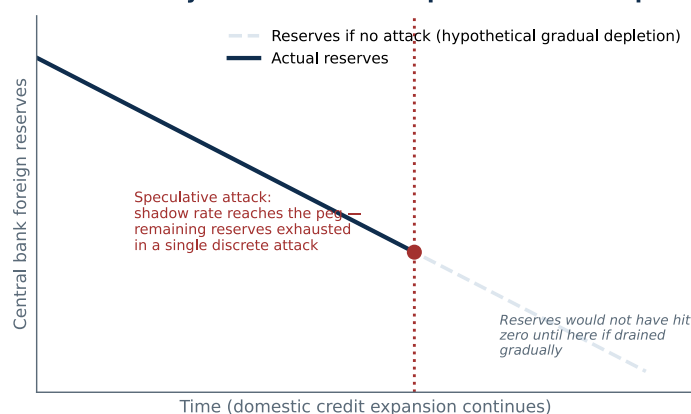


Fig. 3 — Reserves decline steadily as domestic credit expansion continues. Rational speculators attack the currency once the shadow exchange rate reaches the fixed peg — exhausting remaining reserves in one discrete attack, well before reserves would have hit zero under gradual depletion alone.

Why the regime doesn't simply collapse when reserves literally hit zero: the critical insight of Krugman's model is that **rational, forward-looking speculators will not wait for reserves to be gradually exhausted to zero**. If they did, the collapse would be perfectly predictable in advance, creating an anticipated, discrete jump in the exchange rate at a known future date — which would itself be an arbitrage opportunity that any rational investor would exploit *before* that date, not on it. Instead, speculators compare the fixed peg to the "**shadow exchange rate**" — the exchange rate that would prevail *if* the peg were abandoned right now and the currency were allowed to float, given the level of domestic credit and fundamentals at that moment. As domestic credit expansion continues, this shadow rate depreciates over time (moving further from the fixed peg).

The collapse point: the fixed exchange rate regime collapses — via a sudden, discrete speculative attack — at the exact moment the **shadow exchange rate reaches the level of the fixed peg**. At that instant, speculators have no further incentive to wait: holding the currency a moment longer offers no advantage, since the peg is now exactly as attractive as the freely floating alternative, so a rational attack exhausts the central bank's **remaining** reserves in a single, sudden burst well *before* those reserves would have been drawn down to literally zero by the gradual, ongoing credit-driven depletion alone. The regime collapses earlier, and more abruptly, than a naive "reserves run out" story would predict — but at a point that is itself fully determined by the pace of the underlying domestic credit expansion, not by arbitrary market sentiment.

Mark Allocation — Part (c) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Correctly describes the fiscal-deficit/domestic-credit-expansion setup and the resulting gradual reserve depletion AND correctly explains the "shadow exchange rate" concept AND correctly identifies the collapse point as the moment the shadow rate reaches the peg, with the reasoning for why rational speculators attack at that specific point (not waiting for literal reserve exhaustion) explained.
21-30	21-30	Correctly describes the reserve-depletion setup and that a speculative attack occurs before reserves hit zero, but the shadow-exchange-rate mechanism and the precise collapse condition are not clearly explained.
11-20	11-20	General description of a currency crisis (declining reserves, speculative attack) without the specific first-generation model mechanism or collapse condition.
0-10	0-10	No credible description of the model given.

Question A3 Total: 100 Marks